Scale-Correspondence in Event Density: Architectural Invariants Across Magnitudes

Allen Proxmire
March 2026

Abstract
Event Density (ED) is a scale-free ontology. Its primitives—micro-events, ED gradients, participation, and threshold structure—do not depend on size, mechanism, or physical substrate. As a result, the same architectural forms recur across magnitudes: pockets, channels, saddles, divergence zones, stationary points, and threshold transitions appear in systems ranging from nanometer-scale cavities to galaxy-scale mass sheets. This note introduces the ED scale-correspondence program, which identifies these recurring structures and places disparate systems into parallel architectural form. Scale-correspondence is not analogy but identity: when two systems share the same ED geometry, they exhibit the same flow behavior, stability conditions, and structural signatures, regardless of their underlying physics. The purpose of this note is to articulate why ED predicts such recurrences, to define the architectural vocabulary that appears across scales, and to outline the method by which ED-SC overlays extract scale-free rules. The result is a unified picture of becoming in which quantum behavior, classical structure, gravitational curvature, and cosmological evolution are expressions of the same relational substrate.

1. Introduction: ED as a Scale-Free Ontology

Event Density begins from a single ontological commitment: becoming is primitive. The universe is not built from objects, fields, or states, but from micro-events—the discrete acts through which reality continually renews itself. ED is the measurable structure of this activity: the local rate at which micro-events accumulate, stabilize, and form persistent patterns. Geometry, matter, information, and classicality are not fundamental ingredients. They are the macroscopic shadows of how becoming organizes itself across scales.

Because ED is not tied to any particular physical mechanism, substrate, or domain, its structures are architectural rather than material. ED gradients define flows; flows define curvature; curvature defines the macroscopic geometry we interpret as spacetime. Participation bandwidth determines whether a system individuates or remains quantum. ED thresholds determine whether structure stabilizes or dissolves. These architectural features arise from the same primitives—micro-events, ED gradients, and participation—regardless of the scale at which they appear.

This architectural character has a profound consequence: ED structures recur across magnitudes. The same patterns—pockets, saddles, channels, divergence zones, stationary points—appear in nanometer-scale cavities, mesoscopic resonators, astrophysical mass sheets, and cosmological flows. These recurrences are not coincidences or metaphors. They are the natural expression of an ontology whose primitives do not depend on size. When the geometry of becoming is similar, the macroscopic behavior is similar, even when the underlying physics differs completely.

The purpose of this note is to articulate this scale-correspondence explicitly. The ED-SC series identifies pairs of systems that share the same ED geometry despite living in different physical regimes. By placing these systems into parallel structure, the ED-SC overlays reveal the scale-free invariants that govern them. These invariants are the architectural signatures of ED: the forms that persist when matter, forces, and mechanisms change, but the geometry of becoming remains the same.

This note introduces the conceptual foundations of scale-correspondence, explains why ED predicts it, and outlines the architectural vocabulary that recurs across magnitudes. It prepares the reader to understand the ED-SC overlays as more than comparisons—they are demonstrations of ED's unifying power. When the same ED structure appears in systems separated by twenty orders of magnitude, the ontology gains credibility. Elegance becomes evidence.

2. Why Scale-Correspondence Exists in Event Density

Event Density is scale-free because its primitives are scale-free. Micro-events do not come in sizes; participation does not depend on units; ED gradients do not care whether the system is nanometers wide or millions of light-years across. The same architectural rules govern how becoming organizes itself wherever ED varies. Scale-correspondence is therefore not an accident or a curiosity. It is the natural consequence of an ontology whose fundamental structures are relational rather than material.

2.1 ED gradients are geometric, not material

In ED, the primary structure is the flow of becoming. ED gradients determine how rapidly micro-events accumulate from place to place, and this differential rate induces a natural flow field. Curvature is the macroscopic imprint of this flow. Because ED gradients describe how becoming changes, not what matter is present, the same gradient structures can arise in systems with completely different physical mechanisms. A Casimir cavity and a cosmic sheet share no material similarity, yet both produce a planar ED channel with a suppressed direction because the geometry of becoming is the same.

2.2 Micro-events scale, but ED structures do not

Micro-events are discrete acts of becoming. Their production rate varies with local ED, but the shape of the ED landscape—the pattern of gradients, channels, saddles, and divergence zones—does not depend on the absolute number of micro-events involved. A nanometer-scale cavity may involve trillions of micro-events per second; a galaxy sheet may involve vastly more. But the architectural form of the ED channel is identical. ED structures are defined by relations, not magnitudes.

2.3 Participation and curvature are relational

Participation bandwidth determines how richly two regions integrate each other's micro-events. Curvature determines how flow lines converge or diverge. Both are relational quantities: they describe how one region of becoming relates to another. Relational structures do not depend on scale. A thin film, a convection cell, a galaxy filament, and a cosmic void can all instantiate the same participation geometry because the underlying relational pattern is the same.

2.4 ED thresholds create universal regimes

ED thresholds determine whether a system can individuate, stabilize gradients, form curvature, or sustain horizons. These thresholds are structural, not physical. A system becomes quantum when ED is too sparse to maintain internal gradients; it becomes classical when ED is thick enough to stabilize them. These transitions occur at different physical scales in different systems, but the architectural logic is identical. Thresholds are universal because they arise from the same primitive: the capacity of becoming to sustain structure.

2.5 Scale-free behavior as evidence

When the same ED geometry appears in systems separated by twenty orders of magnitude, this recurrence is not a metaphor. It is a diagnostic. Theories that unify physics do so by revealing structures that survive changes in scale. Maxwell's equations, the Schrödinger equation, and general relativity all exhibit this property. ED joins this

lineage by showing that the architecture of becoming—its channels, saddles, pockets, and divergence zones—reappears wherever ED gradients take similar forms. The elegance of this recurrence is itself evidence that ED is describing something real.

2.6 Architectural invariance as the core of ED-SC

The ED-SC program exists because ED predicts that the same architectural forms should appear across magnitudes. When two systems share the same ED geometry, their flows, stationary points, and stability conditions will match, even if their physics does not. ED-SC overlays make this explicit by placing such systems into parallel structure. Scale-correspondence is therefore not an add-on to ED. It is one of its deepest consequences: a sign that the ontology is architectural, relational, and scale-free at its core.

3. The Architectural Vocabulary of ED Across Scales

Event Density reveals a small set of structural forms that recur wherever becoming organizes itself. These forms are not tied to any particular physical mechanism—electromagnetism, gravity, quantum behavior, fluid flow, or cosmological evolution. They arise whenever ED gradients shape participation, and because ED is relational rather than material, the same vocabulary applies from nanometers to megaparsecs. This section introduces the architectural motifs that appear repeatedly in ED-SC overlays and across the ED program as a whole.

3.1 ED Pockets

An ED pocket is a region where becoming is locally suppressed or enhanced relative to its surroundings. Pockets can be low-ED (as in Casimir cavities) or high-ED (as in cosmic sheets), but the polarity is secondary. What matters is the contrast: a pocket is defined by a sharp ED gradient that creates a distinct participation environment. Pockets act as attractors or repellors for flow, depending on whether they represent a valley or a hill in the ED landscape. Their defining feature is relative ED, not absolute magnitude.

3.2 Channels

Channels are participation pathways carved by geometry and ED gradients. They guide the flow of becoming along preferred directions while suppressing motion along others. Channels appear in nanoscale cavities, optical resonators, thin films, galaxy sheets, and cosmic filaments. A channel is defined by three features: a dominant direction of flow, one or more suppressed directions, and a gradient structure that stabilizes the pathway. Channels are the backbone of ED-SC correspondences because they reveal the underlying architecture of participation.

3.3 Saddles

A saddle is a mixed-curvature structure: convergent along one axis, divergent along another. Saddles arise whenever competing ED gradients meet. They appear in gravitational potentials, Casimir cavities, thin-film instabilities, and the Local Group mass sheet. Saddles are structurally important because they create stationary points—locations where flow is neither purely inward nor outward. In ED-SC overlays, saddles often serve as the central invariant: the same saddle geometry can govern systems separated by many orders of magnitude.

3.4 Divergence Zones

Divergence zones are regions where flow cannot accumulate. They occur where ED is too low to sustain structure or where gradients push becoming outward along all available directions. Cosmic voids are divergence zones on cosmological scales; off-plane regions in mass sheets are divergence zones on galactic scales; forbidden separation ranges in Casimir systems are divergence zones on nanoscale. Divergence zones mark the boundaries of channels and pockets. They are the “open directions” in ED-SC correspondences.

3.5 Stationary Points

A stationary point is where competing ED gradients cancel. It is the location of minimal distortion in the flow of becoming. Stationary points determine equilibrium separations in Casimir cavities, stable orbital radii in gravitational systems, and the thickness of thin films. They are the architectural signature of balanced participation. In ED-SC overlays, stationary points often reveal the deepest correspondence: two systems with different physics but identical gradient cancellation.

3.6 Thresholds

Thresholds are ED levels at which qualitative changes in structure occur. When ED falls below the individuation threshold, quantum behavior emerges. When ED rises above the curvature threshold, gravitational structure forms. When ED gradients become too steep to reconcile, horizons appear. Thresholds are universal because they arise from the same primitive: the capacity of becoming to sustain structure. They define the regimes in which ED pockets, channels, saddles, and divergence zones can exist.

These architectural motifs form the vocabulary of ED across scales. They are the recurring shapes of becoming—structures that appear whenever ED gradients organize flow, regardless of the physical substrate. The ED-SC program identifies these motifs in disparate systems and reveals the scale-free rules that govern them.

4. How ED Structures Recur Across Magnitudes

Event Density produces a small set of architectural forms—pockets, channels, saddles, divergence zones, stationary points—that recur wherever becoming organizes itself. These forms appear in systems governed by different forces, different equations, and different physical substrates. Their recurrence is not a coincidence. It is the signature of an ontology whose primitives are relational and whose geometry emerges from ED gradients rather than from matter or mechanism. This section surveys how these structures manifest across four broad regimes: nanoscopic, mesoscopic, astrophysical, and cosmological.

4.1 Nanometer-Scale Systems

At nanoscales, ED gradients are shaped by electromagnetic suppression, boundary conditions, and quantum participation limits. Despite the microscopic setting, the same architectural motifs appear.

Casimir cavities form low-ED pockets where electromagnetic modes are suppressed. The cavity thickness is a stationary point defined by competing ED gradients—Casimir lowering and electrostatic raising.

Quantum wells and thin films create planar channels with one suppressed direction, producing saddle-like ED structures that guide participation.

Optical resonators exhibit stationary points and divergence zones where mode structure forces ED gradients into stable configurations.

Quantum interference platforms reveal ED channels shaped by participation bandwidth and relational timing.

These systems demonstrate that ED structures can arise even when the underlying physics is dominated by quantum behavior and boundary-condition effects.

4.2 Mesoscopic Systems

At intermediate scales—microns to meters—ED gradients are shaped by mechanical, thermal, and fluid-dynamic constraints. Yet the same architectural vocabulary persists.

Fluid convection cells form channels and divergence zones as ED gradients (here expressed through thermal participation) organize flow.

Mechanical resonators exhibit stationary points where restoring forces balance inertial participation.

Biological transport networks form channels and pockets as ED gradients shape efficient participation pathways.

Optical cavities and waveguides create planar channels with suppressed directions, mirroring nanoscale and astrophysical structures.

Mesoscopic systems reveal that ED structures are not tied to quantum or gravitational regimes. They arise whenever participation is constrained by geometry.

4.3 Astrophysical Systems

At galactic and intergalactic scales, ED gradients manifest as curvature. Yet the same motifs reappear.

Galaxy sheets form high-ED planar channels with shallow in-plane gradients and steep off-plane divergence zones.

Filaments act as ED channels that guide matter flow along preferred directions.

Voids are divergence zones where ED is too low to sustain structure.

Orbital basins contain stationary points where gravitational and inertial contributions cancel.

Curvature saddles arise wherever competing mass distributions create mixed convergence and divergence.

These structures mirror the same architectural forms seen in nanocavities and thin films, despite the enormous difference in scale and physical mechanism.

4.4 Cosmological Systems

At the largest scales, ED gradients shape the evolution of the universe itself. The same motifs appear in their most expansive form.

- The cosmic web is a network of ED channels (filaments), pockets (clusters), and divergence zones (voids).
- Inflationary smoothing flattens ED gradients, creating uniform regions analogous to low-ED pockets.
- Horizon formation occurs when ED gradients become too steep to reconcile, producing structural discontinuities.
- Late-time ED diffusion creates large-scale divergence zones as becoming thins and gradients flatten.

Cosmology reveals the same architectural vocabulary that appears in quantum systems, fluid flows, and galaxy sheets. The scale changes; the geometry does not.

4.5 The Same Architectural Forms Across All Regimes

Across these four regimes, the same ED structures recur:

- Pockets — regions of suppressed or enhanced becoming
- Channels — participation pathways shaped by geometry
- Saddles — mixed-curvature structures from competing gradients
- Divergence zones — regions where flow cannot accumulate
- Stationary points — locations where gradients cancel
- Thresholds — ED levels where qualitative behavior changes

These forms are the architectural invariants of ED. Their recurrence across magnitudes is the foundation of the ED-SC program: the recognition that systems with different physics can share the same ED geometry, and therefore the same flow behavior, stability conditions, and structural signatures.

5. The ED-SC Method: How Scale-Correspondence Is Identified

Scale-correspondence is not a metaphorical exercise. It is a disciplined method for extracting architectural invariants from systems that differ in scale, mechanism, and physical domain. The ED-SC method takes two systems that appear unrelated and reveals the shared ED geometry that governs both. This section outlines the steps of that method, each of which mirrors the interpretive moves used throughout the ED Foundations papers.

5.1 Extract the ED Channel

Every ED-SC analysis begins by identifying the ED channel in each system: the participation pathway defined by geometry and ED gradients. A channel is not a physical conduit but a structural alignment of becoming. It is the direction along which micro-events integrate most efficiently. In practice, this means identifying:

- the dominant direction of flow
- the suppressed directions
- the gradient structure that stabilizes the pathway

Once the ED channel is identified in both systems, the first layer of correspondence becomes visible.

5.2 Identify Suppressed and Open Directions

ED channels are defined as much by what they allow as by what they forbid. A suppressed direction is one in which participation is constrained—by geometry, boundary conditions, or ED gradients. An open direction is one in which flow is permitted or even encouraged. The pattern of suppression and openness is often the clearest indicator of architectural identity. Two systems with the same pattern of suppressed and open directions are likely instantiating the same ED structure, even if their physics differ.

5.3 Identify Competing ED Contributions

Many ED structures arise from competing gradients. A cavity thickness, an orbital radius, a stable separation, or a saddle point emerges when two or more ED contributions balance. These contributions may come from:

- Casimir suppression vs. electrostatic raising
- gravitational attraction vs. sheet curvature
- pressure vs. tension
- thermal gradients vs. mechanical constraints

The ED-SC method identifies these contributions in each system and examines how they combine. When two systems exhibit the same pattern of competition, the correspondence deepens.

5.4 Identify the Curvature Structure

Curvature is the macroscopic geometry of ED flow. Once the ED channel and competing gradients are identified, the curvature structure becomes clear. The ED-SC method classifies the structure as:

- a pocket (valley or hill)
- a channel (elongated basin)
- a saddle (mixed curvature)
- a ridge (suppressed direction with outward flow)
- a divergence zone (flow cannot accumulate)

Two systems that share the same curvature structure are expressing the same ED geometry, regardless of scale.

5.5 Identify the Stationary Point

A stationary point is where competing ED gradients cancel. It is the location of minimal distortion in the flow of becoming. Stationary points determine:

- cavity thickness
- equilibrium separations
- stable orbital radii
- flow compensation points

In ED-SC overlays, stationary points often reveal the deepest correspondence. When two systems stabilize at structurally identical stationary points, the architectural identity becomes unmistakable.

5.6 Align the Structures Across Systems

Once the ED channel, suppressed and open directions, competing gradients, curvature structure, and stationary point are identified, the ED-SC method places the two systems into parallel structure. This alignment is expressed through a correspondence table, where each row reveals the same ED feature instantiated at two scales. The table is not a comparison; it is an extraction tool. It reveals the scale-free rule that governs both systems.

5.7 Extract the Scale-Free Rule

The final step is to articulate the architectural invariant: the ED rule that both systems instantiate. This rule is expressed in the same style as the ED Foundations papers—clean, structural, and inevitable. The rule is the heart of the ED-SC overlay. It is the statement that survives when scale, mechanism, and physical substrate are stripped away.

Table 1. Scale-Correspondence between quantum and galactic/cosmological phenomena

ED-I Topic (Micro-Scale)	Cosmological Analogue (Macro-Scale)
001 — Superconductivity	Coherent, low-resistance flow along cosmic filaments
002 — Entanglement	Long-range synchronized behavior between distant voids/filaments
003 — Spin	Galaxy spin alignment across filaments and void boundaries
004 — Quantum–Classical Boundary	Transition zones between void interiors and filament nodes
005 — Magnetism	Directional flow alignment along large-scale filaments
006 — Fields & Forces	Gravitational potentials as tension gradients shaping cosmic flow
007 — Double-Slit	Void–filament bifurcation and interference-like density patterns
008 — Higgs / Mass Generation	Mass-dependent drift in void expansion and filament anchoring
009 — Neutrino Oscillation	Galaxy drift across curvature basins with regime transitions
010 — Gravity Waves	Void "breathing modes" and large-scale curvature oscillations
011 — Baryogenesis	Asymmetric void expansion and anisotropic filament formation
012/018 — Photonics	Photon-like curvature packets moving through void/filament geometry
013 — Quantum Information	Cosmic web evolution as curvature-encoded information processing
014 — Topological Effects	Topologically protected filaments and void walls
015 — Microresonators	Galaxy groups orbiting within curvature wells (resonant basins)
016 — Casimir Self-Assembly	Filament formation from void-driven curvature imbalance
017 — Time-Duration Symmetry	Phase-locked expansion rates across connected voids
019 — Mass Gap	Minimum density thresholds for filament formation
020 — Spin–Momentum Locking	Galaxy spin–flow alignment along filament channels
021 — Chiral Phonons	Chiral galaxy streams around void boundaries
022 — Local Group Anisotropic Mass Structure	Direct cosmological case: asymmetric curvature around the Local Void

6. Case Study: The First Scale-Correspondence Overlay

The first scale-correspondence overlay demonstrates how two systems that appear unrelated—one cosmological, one nanoscale—instantiate the same ED geometry. This case study serves as the template for all future ED-SC analyses. It shows how to extract the ED channel, identify suppressed and open directions, locate competing gradients, and reveal the stationary point that defines the shared architectural rule. The systems differ in scale by more than twenty orders of magnitude, yet their ED structures align cleanly once the material details are stripped away.

6.1 The Local Group Mass Sheet

The reconstruction of the matter distribution around the Local Group reveals a vast, flattened sheet extending tens of millions of light-years. Above and below this sheet lie large, nearly empty regions—divergence zones where ED is too low to sustain structure. Within the sheet, ED gradients are shallow, producing a planar channel along which galaxies drift outward rather than fall inward. The Local Group sits in a curvature saddle, shaped by the competition between inward curvature from the Milky Way–Andromeda pair and outward curvature from the extended mass sheet. This saddle geometry defines the flow: divergence along the plane, suppression perpendicular to it, and a stationary point where the gradients balance.

6.2 The Golden Casimir Cavity

At the nanoscale, gold flakes suspended in salt water form nanometer-scale cavities above a gold substrate. These cavities are low-ED pockets created by the suppression of electromagnetic modes between the surfaces. Two competing ED contributions shape the cavity: Casimir suppression lowers ED, while electrostatic interactions raise it. The flake stabilizes at the stationary point where these gradients cancel. Lateral motion is suppressed by the plate geometry, while vertical motion is guided by the ED channel formed between the surfaces. The result is a saddle-like ED dent whose curvature is visible through the cavity's optical color.

6.3 The Shared ED Geometry

Despite their differences, both systems instantiate the same architectural form:

- a planar channel with one suppressed direction
- two open directions along which flow diverges
- competing ED gradients that define the structure
- a curvature saddle created by mixed convergence and divergence
- a stationary point where the gradients cancel
- flow behavior governed by the same ED logic: drift along the plane, confinement along the suppressed axis

The Local Group sheet and the Casimir cavity differ in scale, mechanism, and physical substrate, but their ED geometries are identical. This identity is what the ED-SC method reveals.

6.4 The Scale-Free Rule

From this correspondence, the architectural invariant emerges:

A planar geometry with one suppressed direction and two competing ED contributions produces a saddle-shaped channel whose flows diverge along the open directions and settle at a stationary point where the gradients cancel.

This rule is not tied to gravity, electromagnetism, or any specific physical mechanism. It is a structural consequence of how ED gradients shape the flow of becoming. The same rule governs nanometer-scale cavities and galaxy-scale mass sheets because ED is a scale-free ontology.

6.5 Why This Case Matters

This first overlay demonstrates the core insight of the ED-SC program: when the geometry of becoming matches, the behavior matches, regardless of scale. It shows that ED structures are architectural forms that recur across magnitudes, and that identifying these forms reveals deep unity across physics. The Local Group–Casimir correspondence is not an isolated curiosity. It is the first entry in a growing atlas of scale-free ED geometries.

7. The Significance of Scale-Correspondence

Scale-correspondence is more than an interpretive curiosity. It is one of the strongest indicators that Event Density is uncovering a real architectural substrate beneath the diverse phenomena of physics. When the same ED structures appear in systems separated by twenty, thirty, or even forty orders of magnitude, this recurrence is not decorative. It is diagnostic. It signals that ED is identifying the geometry of becoming—a structure that persists when matter, forces, and mechanisms change, but the relational architecture remains the same.

7.1 Elegance as Evidence

In physics, elegance is not an aesthetic preference. It is a sign that a theory is compressing complexity into a smaller set of generative principles. Maxwell's equations unify electricity, magnetism, and light. The Schrödinger equation unifies atomic spectra, chemical bonding, and interference. General relativity unifies gravity, geometry, and inertial motion. In each case, the theory's credibility grows when it explains more with less.

ED exhibits this same signature. The fact that the same ED geometry appears in nanocavities, thin films, galaxy sheets, and cosmic voids is not a coincidence. It is evidence that ED is describing something real—an underlying architecture that does not depend on scale. When a cavity thickness and a galaxy drift arise from the same saddle structure, the ontology gains credibility. The world becomes simpler, not more complicated.

7.2 ED as a Unifying Architecture

ED does not unify physics by adding new mechanisms. It unifies physics by revealing that many mechanisms are shadows of the same architectural form. Quantum behavior arises when ED is too sparse to individuate. Curvature arises when ED gradients shape flow. Information arises from the pattern of becoming; entropy from its diffusion. Cosmology is the long-scale evolution of ED as gradients smooth and stabilize.

Scale-correspondence shows that these domains are not separate. They are different expressions of the same relational substrate. A Casimir cavity and a cosmic sheet are not analogous—they are architecturally identical. Their physics differ; their ED geometry does not.

7.3 Predictive Power

Scale-correspondence is not retrospective. It is predictive. Once ED identifies a structural form—such as a planar saddle with one suppressed direction—it becomes possible to search for that form in other systems. The ED-SC method suggests new correspondences:

- thin films ↔ galaxy walls
- nanowires ↔ cosmic filaments
- optical cavities ↔ gravitational wells
- convection cells ↔ cosmic void flows
- quantum wells ↔ orbital basins

These predictions arise from geometry, not mechanism. If the ED channel matches, the behavior should match. This gives ED a new kind of predictive power: the ability to anticipate structural behavior across domains that traditional physics treats as unrelated.

7.4 A New Way to Read Physical Systems

Scale-correspondence reframes how physical systems are understood. Instead of asking:

- What forces act here?
- What equations govern this domain?
- What material properties matter?

ED asks:

- What is the geometry of becoming?
- Where are the gradients?
- What channels, pockets, saddles, and divergence zones shape participation?
- What stationary points define stability?

This shift allows systems to be compared architecturally rather than mechanistically. It reveals unity where physics traditionally sees fragmentation. It shows that the same structural logic governs quantum wells, fluid flows, galaxy sheets, and cosmic evolution.

7.5 The Role of ED-SC in the ED Program

The ED-SC series is the natural extension of the ED Foundations papers. The foundations establish the ontology: micro-events, ED gradients, participation, curvature, thresholds. The ED-SC series shows how these primitives generate recurring structures across magnitudes. It is the bridge between ED's conceptual architecture and the physical world's diversity.

Scale-correspondence is therefore not an optional add-on. It is the empirical face of ED's architectural commitments. It demonstrates that ED is not merely a reinterpretation of known physics but a unifying framework capable of revealing deep structural invariants.

8. The Road Ahead for the Scale-Correspondence Program

The Scale-Correspondence program is the natural continuation of the ED Foundations work. The foundations establish the ontology—micro-events, ED gradients, participation, curvature, thresholds. Scale-correspondence shows how these primitives generate recurring structures across magnitudes. The road ahead is therefore not a matter of expanding the theory, but of mapping its architectural consequences across the physical world. This section outlines how the ED-SC series will grow, what it aims to reveal, and how it will shape the broader ED program.

8.1 Building a Canon of Architectural Invariants

The first overlay demonstrated that a nanoscale cavity and a galaxy-scale mass sheet instantiate the same ED geometry. Future overlays will extend this catalog. Each ED-SC entry will identify a new architectural invariant—an ED structure that recurs across systems with different physics and different scales. Over time, these overlays will form a canon of scale-free ED geometries, a reference library of the universe's recurring structural motifs.

8.2 Toward a Scale-Free ED Atlas

As the catalog grows, the ED-SC series will evolve into a scale-free atlas: a map of the architectural forms that appear wherever ED gradients organize becoming. This atlas will not be organized by physical domain—quantum, fluid, gravitational, cosmological—but by geometry:

- planar channels
- curvature saddles
- ED pockets
- divergence zones
- stationary points
- threshold transitions

Systems will be grouped by the structures they instantiate, not by the mechanisms that produce them. This reframing will reveal connections that traditional physics obscures.

8.3 Identifying New Correspondences
The ED-SC method is predictive. Once an architectural form is identified, it becomes possible to search for it in other systems. The road ahead includes identifying correspondences such as:

- thin films ↔ galaxy walls
- nanowires ↔ cosmic filaments
- optical cavities ↔ gravitational wells
- convection cells ↔ cosmic void flows
- quantum wells ↔ orbital basins

These correspondences are not analogies. They are architectural identities. The ED-SC program will systematically uncover them.

8.4 Implications for Theory and Experiment
Scale-correspondence has implications beyond interpretation. It suggests new experimental platforms for probing ED structures. If a nanoscale system instantiates the same ED geometry as a cosmological one, the nanoscale system becomes a laboratory proxy for the larger structure. Conversely, astrophysical observations can reveal architectural features that guide nanoscale design. The ED-SC program opens a two-way bridge between scales.

8.5 ED as a Discovery Engine
The ED-SC series transforms ED from a conceptual framework into a discovery engine. By identifying architectural invariants, ED-SC reveals hidden structure in systems that appear unrelated. It suggests new experiments, new interpretations, and new ways of reading physical systems. It shows that the universe is not a collection of disconnected domains but a single architectural landscape shaped by becoming.

8.6 The Long-Term Vision
The long-term goal of the ED-SC program is to articulate a unified architectural picture of the universe—one in which quantum behavior, classical structure, gravitational curvature, information flow, and cosmic evolution are all expressions of the same relational substrate. Scale-correspondence is the key to this vision. It shows that ED is not merely a reinterpretation of known physics but a framework capable of revealing deep structural unity across magnitudes.

9. Conclusion
Event Density begins with a single primitive—becoming—and from that primitive, a remarkably small set of architectural forms emerges. These forms are not tied to matter, forces, or mechanisms. They arise wherever ED gradients shape participation, and because ED is relational rather than material, these forms recur across

magnitudes. A nanometer-scale cavity and a galaxy-scale mass sheet can instantiate the same ED geometry because both are expressions of the same underlying structure: the flow of becoming through a landscape of gradients.

Scale-correspondence makes this unity visible. It shows that pockets, channels, saddles, divergence zones, stationary points, and thresholds are not domain-specific constructs. They are the universal shapes of ED flow. When these shapes appear in systems separated by twenty or thirty orders of magnitude, the recurrence is not metaphorical. It is architectural. It is the signature of an ontology that is scale-free at its core.

The ED-SC program formalizes this insight. By placing disparate systems into parallel structure, it reveals the invariants that survive when scale, mechanism, and physical substrate are stripped away. These invariants are the scale-free rules of ED—the generative principles that govern quantum behavior, classical structure, gravitational curvature, information flow, and cosmic evolution. The first overlay demonstrates this explicitly: a Casimir cavity and a cosmic sheet share the same ED geometry. Many more such correspondences remain to be uncovered.

The significance of this is twofold. First, it strengthens the ED ontology. When a theory explains more with less—when it compresses diverse phenomena into a small set of architectural forms—its credibility grows. Elegance becomes evidence. Second, it transforms ED into a discovery engine. Scale-correspondence suggests new experiments, new interpretations, and new ways of reading physical systems. It reveals unity where physics traditionally sees fragmentation.

The road ahead is to build a scale-free atlas of ED structures: a catalog of the universe's recurring architectural motifs. As this atlas grows, the ED-SC series will become a map of how becoming organizes itself across magnitudes. It will show that the universe is not a collection of disconnected domains but a single architectural landscape shaped by ED gradients and participation.

The task now is to explore that landscape.